Xinyi Yan

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Education

University of Texas at Austin, PhD candidate in EEB (2020 – present; expected spring 2026)

- Advisors: Dr. Caroline Farrior & Dr. Amelia Wolf
- Portfolio Program in Applied Statistical Modeling

University of California, Los Angeles, B.S. (2016 – 2020)

- Majors in: Applied Mathematics & Biology (Departmental Highest Honors)
- Magna Cum Laude

Peer Reviewed Publications

Zou, H. X., **Yan, X.**, & Rudolf, V. H. (2024). Time-dependent Interaction Modification Generated from Plant-soil Feedback. *Ecology Letters.* http://doi.org/10.1111/ele.14432

McGuire, R. M.*, Hayashi, K. T.*, **Yan, X.***, Carita Vaz, M., Cinoğlu, D., Cowen, M. C., ... & Kandlikar, G. S. (2022). EcoEvoApps: Interactive Apps for Teaching Theoretical Models in Ecology and Evolutionary Biology. *Ecology and Evolution.* http://dx.doi.org/10.1002/ece3.9556 *equal contributors.

Yan, X.*, Levine, J. M., & Kandlikar, G. S.* (2022). A quantitative synthesis of soil microbial effects on plant species coexistence. *Proceedings of the National Academy of Sciences,* 119(22), e2122088119. https://doi.org/10.1073/pnas.2122088119 *equal contributors.

Kandlikar, G. S., **Yan, X.**, Levine, J. M., & Kraft, N. J. (2021). Soil Microbes Generate Stronger Fitness Differences than Stabilization among California Annual Plants. *The American Naturalist*, 197(1), E30-E39. https://doi.org/10.1086/711662

Kandlikar, G. S., Johnson, C. A., **Yan, X.**, Kraft, N. J., & Levine, J. M. (2019). Winning and losing with microbes: how microbially mediated fitness differences influence plant diversity. *Ecology Letters,* 22(8), 1178-1191. https://doi.org/10.1111/ele.13280

Manuscripts Available Upon Request

Yan, X., Bowman, E.A., Ortiz, S.K., Cinoğlu, D., Farrior, C.E., & Wolf, A.A (2025). Plant diversity and water regime have distinct effects on soil fungal communities that are linked to plant productivity. *bioRxiv.* https://doi.org/10.1101/2025.10.10.680786

Ita, R. E., **Yan, X.**, Krishnadas, M., Ke, P. J., Harpole, W. S., Barabas, G., V-T, Y., & Kandlikar, G. (2025). Rethinking plant community assembly through the lens of soil microbial regulation. *Under revision at International Journal of Plant Sciences.*

Senthilnathan, A., Ke, P. J., **Yan, X**., Crawford, C., & D'Andrea, R. (2025). Plants, soil, and coexistence: an alternative approach to modelling plant-soil feedback. *Submitted to Plant and Soil.*

Krishnadas, M., **Yan, X.**, Ke, P.J., Barabas, G., Harpole, W.S., Corrales, A., Tiruvaimozhi, Y.V., & Kandlikar, G. (2025). Microbially mediated coexistence of plant species predicted by root traits. *Manuscript available.*

Fellowships & Awards

2026 Spring Dissertation Writing Fellowships, UT Austin ($17k)
2025 Santa Fe Institute Complexity System Summer School Tuition Scholarship ($2.9k)
2024 University Graduate Continuing Fellowship, UT Austin ($36k)
2023 Stengl-Wyer Fellowship, UT Austin ($36k for stipend + $2k for research)
2021-2023 Research Award by the Dept. of Integrative Biology, UT Austin ($7.5k accumulated)
2022 Blattstein Fellowship, UT Austin ($3k)
2022 Ann Miller Gonzalez Graduate Research Grants, Native Plant Society of Texas ($2.5k)
2022 Marion Elizabeth Eason Endowed Scholarship, UT Austin ($1.1k)
2020 Recruitment Fellowship, UT Austin ($31k)
2019 CALeDNA Summer Research Internship, UCLA ($3k)
2019 First Place poster at EEB Annual Biology Research Symposiums, UCLA ($100)
2018 Undergraduate Research Fellow, UCLA ($2k)

Invited Working Group Participations

Early career working group at the Synthesis Center (sDiv) of German Center for Integrative Biodiversity Research (iDiv). Soil microbial controls over plant coexistence: synthesizing data and theory to unravel context dependence of plant-soil feedbacks under global changes (sPSF). 2021 – 2024.

Invited Presentation at Conference

YAN, X., Levine, J. M., & Kandlikar, G. Linking theory and global data to evaluate microbial control over plant coexistence. *Organized Oral Session* in *2022 ESA+CSEE Annual Meeting* (August 14-19).

Contributed Presentations at Conferences (since graduate school)

YAN, X., Farrior, C. Microbial competition for plant resources leads to multi-stability in plant coexistence outcomes. In *2025 ESA Annual Meeting* (August 10-15).

YAN, X., Bowman, E. Ortiz, S., Cinoglu, D., Martinez, J., Farrior, C., & Wolf A. The impact of plant diversity and water availability on soil fungal communities, and their subsequent effect on plant productivity. In *2025 ASN Asilomar Conference* (Jan 3-7).

YAN, X., Bowman, E. Ortiz, S., Cinoglu, D., Farrior, C., & Wolf, A. The effect of plant diversity and water availability on soil fungal communities: A field study. In *2024 ESA Annual Meeting* (August 4-9).

YAN, X., Bowman, E. Ortiz, S., Cinoglu, D., Farrior, C., & Wolf, A. The effect of plant diversity and water availability on soil fungal communities: A field study. In *2024 Native Plant Society of Texas Spring Symposium* (March 1).

FARRIOR, C. E., Decker, R. R., Bradly, M. L., Cinoglu, D., & **Yan, X.** Can we predict the evolution of differences among species that allow for their coexistence? In *2022 ESA+CSEE Annual Meeting* (August 14-19).

YAN, X., Levine, J. M., & Kandlikar, G. Quantifying the effect of plant-soil feedbacks on plant coexistence: A meta-analysis of the microbially mediated fitness difference. In *2021 ESA Annual Meeting* (August 2-5).

YAN, **X.,** Kandlikar, G.S., Levine, J. M., & Kraft, J.B. Evaluating the microbial effect on the pairwise and community-wide coexistence of California annual species. In *2020 ESA Annual Meeting* (August 3-6).

KANDLIKAR, G. S., Cowen, M., Hayashi, K., McGuire, R. Vaz, M.C., & **Yan, X.** Interactive web-apps for theoretical ecology active-learning modules. In *2020 ESA Annual Meeting* (August 3-6).

Training & Workshops

2025 Santa Fe Institute Complexity System Summer School (4-weeks introduction to complex behavior in mathematical, physical, living, and social systems, with a group project)

Peer Review

Reviewed and co-reviewed 16 papers for 6 journals: **American Journal of Botany** | **Ecological Monographs** | **Ecology** | **Ecology Letters** | **Journal of Ecology** | **New Phytologist.**

Teaching, Mentoring, & Instructional Design

Co-developer/maintainer, *EcoEvoApps:* (May 2020 – present)

- Develop open-source apps that simulate ecological models, available in an R package and on the web
- Co-first author of a research paper on using *EcoEvoApps* in undergraduate EEB education

Teaching Assistant, University of Texas at Austin (2020F, 2021F, 2025F)

- Courses: Biostatistics (upper division), Introduction to Biology II (lower division), Ecology (upper division)
- Led discussion sections or R computer labs for 40-100 students per semester

Undergraduate Mentorship (May 2024 – May 2025)

- Mentoring two UT undergraduate students on fieldwork, molecular work, experimental design, statistical analysis, and academic writing.
- One student received the UT Undergraduate Research Fellowship ($700) and lead independent research.

Teaching Preparation Series Advanced Certificate, *Center for Teaching & Learning* (UT Spring semester 2023)

- Completed the semester-long workshop to improve my skills and knowledge on teaching and mentorship

Services

Graduate Counsil Representative, *American Society of Naturalists* (Sept. 2022- present)

- Organize events at conferences, design merchandises, and coordinate with the ASN committees to promote DEI and professional developments in graduate students
- Served as a reviewer for the 2023 ASN Student Research Award

Member, Departmental Pre-candidacy Grad Mentoring Working Group (April 2023 – present)

- Organize bi-semester departmental townhall discussions for pre-candidacy students on pre-candidacy activities and milestones

Co-organizer, 'Ecolunch' departmental ecology seminar (Aug. 2022 – June 2023)

- Organized and moderated weekly departmental ecology seminars

Outreach & Science Communication

Outreach Lectures at *Neighborhood Science* & *Science Under the Stars*: (Nov. 8th & 10th 2022)

- Introduce to the public how soil microbes influence plant individual growth, assembly of natural communities, and our daily life

Outreach Lecture at *Botany of the Holidays*, *UCLA Mildred E. Mathias Botanical Garden*: (Dec. 3rd 2020)

- Introduce the plant garlic and its cultural role Chinese winter holidays

Volunteer Docent, *Zilker Botanical Garden*: (Feb. 2022 – present)

- Lead garden tours and educational activities (at garden workshops, Roots & Wings Festival, etc.) for schools and the general public

Volunteer & Docent, *Science Under the Stars*: (Sep. 2022 – Sep. 2025)

- Lead public tours around the Brackenridge Field Laboratory; help with set-up of SUTS talks

Volunteer Docent, *UCLA Mildred E. Mathias Botanical Garden*: (Dec. 2018 – June 2020)

- Lead educational garden tours for schools and the general public